

Technical SEO and On-Page Specification

Astro (SSG) · Payload CMS · Postgres · Cloudflare

Companion to `01-site-strategy-and-content-architecture.md`. That document decides which pages exist. This one decides how they are built, served, marked up, and monitored.

Rev 2 — 29 July 2026. The public-branch model is removed (ADR-004 in the build plan). No branch pages, no branch sitemap shard, no `MovingCompany` / `LocalBusiness` entities, no per-city Google Business Profiles. Structured data carries one `Organization`; every geo page emits `Service` + `areaServed` naming the serviceable city or locality.

1. Rendering and build architecture

Everything indexable is static. No SSR on any URL Googlebot can reach. Astro's default output is what we want; the only server-rendered surfaces are the quote API, the tracking lookup, and search, all of which are noindex or POST-only.

Build pipeline

```
Payload (Postgres)
├─ publish gate query → eligible page manifest (JSON, ~6k rows)
│   └─ Astro getStaticPaths reads the manifest
│       └─ static HTML → Cloudflare Pages/R2
│           └─ sitemap shards generated from the same manifest
```

The manifest is the single source of truth. Sitemaps, internal link graphs, and the route table all derive from it, so they cannot drift apart. This is the structural fix for the incumbent's biggest problem: their sitemap and their actual site are two unrelated artefacts, and 160 internally-linked URL families never made it into the sitemap.

Build time at scale

6,000 pages in Astro is fine, but plan for it before it hurts:

- Fetch once, not per page. Load all Payload data into memory in a single pass, index it in Maps, and have `getStaticPaths` read from memory. Per-page API calls turn a 4-minute build into a 40-minute one.
- Content Layer API for anything file-based; it caches between builds.
- Incremental builds by shard. Split the build by page family so a rate-card change rebuilds only `/routes/*`. Astro doesn't do this natively, so drive it from CI: a change manifest from Payload webhooks determines which shards run.

- Budget: full cold build under 12 minutes, shard rebuild under 2. Past that, engineers stop rebuilding, and stale pages ship.

Publish triggers

Payload afterChange hook → queue → debounce 10 minutes → trigger shard build. Debouncing matters; an ops team editing 40 rate cards should cause one build, not forty.

2. Cloudflare cache contract

The site we audited sits behind Cloudflare and sends `cache-control: no-store, no-cache, must-revalidate` with `cf-cache-status: DYNAMIC` on every page. All 10,000 URLs hit origin on every request. Do not repeat that.

ASSET	CACHE-CONTROL	EDGE TTL
HTML (static pages)	<code>public, max-age=0, s-maxage=86400, stale-while-revalidate=604800</code>	1 day, SWR 7 days
Hashed JS/CSS	<code>public, max-age=31536000, immutable</code>	1 year
Images (hashed)	<code>public, max-age=31536000, immutable</code>	1 year
sitemap*.xml, robots.txt	<code>public, max-age=3600, s-maxage=3600</code>	1 hour
API / quote endpoints	<code>private, no-store</code>	bypass

Purge on deploy by cache tag, not by URL list. Tag HTML responses with their page family (`tag: routes` , `tag: city-mumbai`) so a shard rebuild purges only what changed.

Enable Tiered Cache and Cache Reserve. With SSG output the origin is object storage, so hit ratio should sit above 98%. Alert if it drops below 95%.

Early Hints for the LCP image and the primary font. Cheap, and worth 100–200ms on mobile.

Do not put the quote form behind Bot Fight Mode; it will break legitimate submissions. Use Turnstile on the form instead.

3. The URL contract

This is the most important section in the document.

Every valid URL corresponds to a row in the manifest. There is no route handler that generates a page from an arbitrary path segment.

The audited competitor returns HTTP 200 with `index, follow` for `/zzzfaketown-packers-movers-delhi` , `/home-shifting-in-zzzfaketown` , and `/packers-movers-from-zzzfaketown-to-mumbai` . Their URL space is

infinite. With Astro SSG this failure mode is impossible by default, provided nobody adds a catch-all `[...slug].astro` with a database lookup and a fallback render. Ban that pattern in code review.

Slug rules

- Lowercase, hyphen-separated, ASCII only. Transliterate Devanagari at ingest (नोएडा → noida), store the transliteration in Postgres, never generate it at request time.
- Slugs are immutable once published. Renames create a redirect row; they do not change the URL.
- Enforce uniqueness with a Postgres unique index on `(collection, slug)`. Collisions fail the build, loudly.
- No trailing slash, or always trailing slash. Pick one, set `trailingSlash: 'never'` in `astro.config`, and add a Cloudflare rule to 301 the other form.
- Force lowercase at the edge: a 301 from any uppercase path to its lowercase equivalent. The incumbent serves `/ROHINI-packers-movers-delhi` at 200 with a self-referencing canonical, which is a duplicate-URL generator.

Status code contract

SITUATION	RESPONSE
Valid page	200
Slug changed	301 to new slug
Location no longer serviceable	301 to parent city hub
Page failed the publish gate, never existed	404
Page existed, deliberately retired	410
Pattern-shaped URL with unknown token	404. Never a templated 200.
Typo'd / random path	404 with a real 404 status, not a 200 carrying a noindex tag
Case or trailing-slash variant	301 to canonical form

The 404 page is a static Astro page returned with a genuine 404 status via Cloudflare Pages' `404.html` handling. Verify with `curl -I` in CI, because "renders a 404 page but returns 200" is the single most common soft-404 bug.

Query parameters

No indexable page takes a meaningful query parameter. Filters and sorts are either static paths or client-side. Cloudflare strips known tracking parameters (`utm_*`, `gclid`, `fbclid`, `msclkid`) from the cache key. Canonical tags are always parameter-free.

4. Canonicalisation

- Self-referencing absolute canonical on every indexable page: `https://example.com/path`.

- Canonical is generated from the manifest row, never from Astro.url. Deriving it from the request is how you end up canonicalising uppercase variants to themselves.
- Paginated listings: each page self-canonicalises; rel=prev/next is dead but harmless; do not canonicalise page 2 to page 1.
- One hostname. 301 www → apex (or the reverse), 301 all HTTP → HTTPS, HSTS with `max-age=31536000; includeSubDomains; preload`.
- No hreflang unless we ship regional language versions. If we do, hi-IN, en-IN, and x-default, with full reciprocal annotation and separate URL paths (`/hi/...`), never cookie or IP-based switching.

5. Sitemaps

Generated from the manifest at build time. One index, sharded by page family:

```

/sitemap.xml           index
/sitemaps/core.xml     home, services, corporate
/sitemaps/cities.xml   city hubs
/sitemaps/city-services.xml city x service
/sitemaps/localities-01.xml localities (10k cap per shard)
/sitemaps/routes-01.xml routes
/sitemaps/guides.xml
/sitemaps/reviews.xml

```

Rules:

- `lastmod` is the real `updated_at` from Postgres, in W3C datetime. Not the build timestamp. The audited site sets an identical `lastmod` on all 9,921 URLs, which makes the field worthless.
- Drop `priority` and `changefreq` entirely. Google ignores both. Their site encodes tree depth in priority and gains nothing.
- Only canonical, indexable, 200-returning URLs. A sitemap containing a redirect or a noindex page is a quality signal against us.
- Sharding gives per-family indexation ratios in GSC, which is the single most useful diagnostic we will have.
- CI check: every manifest row appears in exactly one shard, and every sitemap URL resolves to 200 with a self-referencing canonical. Fail the build otherwise.

6. robots.txt

```
User-agent: *
Allow: /
Disallow: /search
Disallow: /quote/thank-you
Disallow: /track/result
Disallow: /*?

Sitemap: https://example.com/sitemap.xml
```

On AI crawlers. The competitor blocks ClaudeBot, GPTBot, Google-Extended, CCBot and meta-externalagent through Cloudflare's managed rules. For a lead-generation business that is a mistake. A growing share of "who should I hire to move to Pune" now happens inside an assistant, and being absent from that surface has no upside. Allow the retrieval crawlers. If there is a policy objection to training use specifically, set `Content-Signal: search=yes, ai-train=no, use=reference` and leave the user agents allowed. Revisit quarterly.

Block only the scrapers that cost bandwidth and give nothing back.

7. Internal linking implementation

Computed in Postgres, materialised into the manifest, rendered by Astro. Never hand-authored, never a mega-menu.

```
-- nearest sibling localities, precomputed into a link table
SELECT a.id AS from_id, b.id AS to_id,
       ST_Distance(a.geo, b.geo) AS d
FROM locations a
JOIN locations b
  ON b.parent_id = a.parent_id
 AND b.id <> a.id
 AND b.is_published
WHERE a.type = 'locality' AND a.is_published
ORDER BY a.id, d
LIMIT 6;
```

Do the same for related routes (shared endpoint, ranked by job volume), same-service-nearby-city, and guide relevance by tag overlap. Materialise as an `internal_links` table refreshed nightly.

Build-time assertion: every published page has at least 3 inbound contextual links, excluding header, footer, and breadcrumbs. Fail the build if any page has fewer. This one check is what prevents the incumbent's 95.8%-orphaned outcome.

- Breadcrumbs on every non-home page, rendered as visible UI and marked up with BreadcrumbList.
- Outbound link budget: 80 per page. City hubs with more than 60 localities paginate.

- Anchor text comes from data fields, rotated by slot position. `{Locality}`, `Movers in {Locality}`, `{Service} in {Locality}` cycling across positions. Never a single repeated exact-match anchor across thousands of pages.

8. Structured data

JSON-LD only, injected from the manifest row. No microdata, no duplicate markup between the two.

PAGE TYPE	TYPES
All	<code>Organization</code> (once, on home, with <code>@id</code> referenced elsewhere), <code>WebSite</code> with <code>SearchAction</code> , <code>BreadcrumbList</code>
City hub / locality	<code>Service</code> with <code>areaServed</code> naming the serviceable city or locality, <code>provider</code> → the <code>Organization @id</code>
City × service	<code>Service</code> + <code>Offer</code> with <code>priceSpecification</code> from the rate card, <code>priceCurrency</code> : INR
Route	<code>Service</code> + <code>Offer</code> ; consider <code>TripDeparture</code> -style properties only if genuinely applicable
Guide	<code>Article</code> with <code>author</code> , <code>datePublished</code> , <code>dateModified</code> , <code>reviewedBy</code>
FAQ blocks	<code>FAQPage</code> only where the Q&A is genuinely on the page and visible
Reviews	<code>Review</code> on individual review items, with <code>author</code> and <code>datePublished</code>

Rules that keep us out of trouble:

- No `MovingCompany` or `LocalBusiness` entities anywhere.** We publish no physical premises, so any local-business entity would be fabricated — exactly the kind of thing that triggers a spam action, and a Google Business Profile suspension risk. One `Organization` carries the legal identity (`legalName`, `GSTIN` as `taxID`, `CIN` as `identifier`, `registeredOffice`); every geo page points at it via `provider` and scopes itself with `areaServed`. CI enforces this (gate G10).
- Never self-serving `AggregateRating` on our own `Organization`. It has been ineligible for rich results since 2019 and it looks like manipulation at 6,000-page scale.
- No `Product` markup. The competitor puts `Product` on every one of its pages for a services business; Google will ignore it at best.
- Prices in `offer` must match visible prices on the page and stay current. Stale price markup is worse than none.

9. On-page template specification

Variables in `{braces}` come from the manifest. Every template must produce output within the character budgets without truncation for the longest real value in the database. Test with the longest slug you actually have, not with "Delhi."

Title patterns

PAGE TYPE	PATTERN	BUDGET
City hub	Packers and Movers in {City} - {Brand}	≤60
City × service	{Service} in {City} - Fixed Quotes, {Brand}	≤60
Locality	Packers and Movers in {Locality}, {City} - {Brand}	≤60
Route	{Origin} to {Destination} Packers and Movers - {Brand}	≤60
Guide	{Headline}	≤60

No phone numbers in titles. No pipe-stuffed keyword chains. The competitor's `Best Home Shifting in Rohini, 9070201040 | OM Packers and movers near me` is 72 characters of which about 20 are useful.

Meta descriptions

Composed from data, not from a single spun sentence. 140–160 characters.

```
{Service} in {Locality} from ₹{price_from}. {job_count_12m} moves completed in {Locality}, {on_time_pct}% on time. Fixed quote, no surprises.
```

If the data is missing, fall back to a shorter description rather than padding. A missing description is better than an identical one on 4,000 pages.

Heading structure

- Exactly one `<h1>`, matching user intent, not the title tag verbatim.
- `<h2>` per major section, in a data-driven order (sections with more data float higher).
- No heading used purely for keyword placement.

Validate every template against the Rich Results Test in CI, not manually.

Section order for templated pages

1. H1 + trust bar (registration, insurance, years operating)
2. Quote widget with location pre-filled
3. Pricing table from the rate card
4. What's included / what's extra
5. Serviceable-area confirmation: the areas we cover in this city, jobs completed there
6. Local operational notes (the section that must be unique or the page doesn't publish)
7. Reviews scoped to this location or route
8. Related links (siblings, services, routes, guides)
9. Scoped FAQs

10. Single closing CTA

Sections 3, 6, and 7 are the ones that carry uniqueness. If any is empty, the gate should already have rejected the page.

10. Core Web Vitals

Targets at p75, mobile, throttled 4G:

METRIC	TARGET	NOTES
LCP	< 1.8s	Usually the hero image or H1
INP	< 150ms	Quote widget is the main risk
CLS	< 0.05	Reserve space for reviews, maps, images
TTFB	< 200ms	Trivial from edge cache; if it isn't, cache config is wrong

Astro-specific implementation

- Zero JS by default. Only the quote widget, the mobile nav, and the review carousel hydrate. Use `client:visible` for anything below the fold, `client:idle` for the nav. Never `client:load` on a below-fold island.
- No React unless a component genuinely needs it. The quote form can be a single Preact or vanilla island; pulling in React for one form costs ~45KB gzipped for nothing.
- Fonts: one family, two weights, self-hosted, woff2, `font-display: swap`, preloaded. No Google Fonts request (extra connection, and a privacy question in some jurisdictions). (*Amended by ADR-002: two families ship — display serif for H1/section heads only, plus the text grotesk — subset to ~52KB total.*)
- Maps: never embed an iframe map on a templated page. Render a static map image and load the interactive map on click. A Google Maps embed on 6,000 pages is a Core Web Vitals disaster.
- Third-party scripts: GTM via Partytown or, better, server-side tagging through Cloudflare Zaraz. No analytics script in the critical path. Meta Pixel and any chat widget load on interaction only.
- CSS: Astro scopes per-component; inline critical CSS, keep total under 40KB.

Images

- Astro `<Image>` / `<Picture>`, AVIF with WebP fallback, explicit width/height on every instance.
- LCP image: `loading="eager"`, `fetchpriority="high"`, preloaded. Everything else `loading="lazy"`.
- Cloudflare Images or Astro's Cloudflare image service for on-the-fly resizing; never ship a 2400px original to a 360px viewport.
- Alt text from data: `{Brand} crew loading a truck in {Locality}, {Month} {Year}`. Descriptive, not keyword-stuffed.
- Image count discipline: the audited competitor renders 641 images on one state hub page. Cap at 25 per page.

11. Indexation control and monitoring

Controls

- `<meta name="robots" content="index, follow">` is the default and does not need to be emitted. Emit `noindex` explicitly only where needed: `/search`, thank-you pages, tracking results, paginated pages beyond page 3 of low-value listings.
- No `nofollow` on internal links. If a link shouldn't pass equity, it shouldn't be there.
- `X-Robots-Tag: noindex` at the edge for the staging hostname, plus HTTP auth. Never rely on `robots.txt` alone to keep staging out of the index.

Monitoring

CADENCE	CHECK	THRESHOLD / ACTION
Per deploy	CI: status codes, canonicals, sitemap parity, structured data, duplication score, inbound-link assertion	Build fails
Daily	Cloudflare cache hit ratio, 5xx rate, origin response time	Alert below 95% hit, above 0.1% 5xx
Weekly	GSC Index Coverage per sitemap shard	Any shard below 60% indexed → pause publishing that family
Weekly	Core Web Vitals from CrUX + RUM	Any metric out of target → triage
Fortnightly	Cloudflare Logpush → crawl analysis: Googlebot hits by page family, crawl waste on 404/301	Rebalance internal links toward under-crawled families
Monthly	Rank tracking by page family, not by vanity keyword	Trend, not absolute position
Monthly	Pages with zero impressions in 90 days	Review, improve, or unpublish

Pull GSC data via the API into Postgres and join it against the manifest. That gives per-page-type indexation and performance in one query, which is the reporting view that actually drives decisions and which almost nobody builds.

Log-based crawl analysis

Cloudflare Logpush to R2, parsed weekly. Look for: Googlebot requests to URLs not in the manifest (means something is generating links we don't know about), crawl budget spent on redirects, and page families being crawled far below their share of the site.

12. Local SEO

Organic and local are separate ranking systems. Because we publish no physical premises, the local surface is deliberately small — and honest:

- **One Google Business Profile only**, for the registered office, with the real address that passes verification. **Never create GBPs for cities where there is no premises**; suspension is likely and the

recovery process is painful, and a fabricated local presence is exactly the trust failure the whole strategy is built against.

- NAP consistency between the GBP, the `Organization` JSON-LD, and the footer on every page. Any mismatch weakens both.
- The GBP links to the `/contact` page, which lists every serviceable city and area.
- Service areas defined on the GBP for the cities we actually cover, matching `areaServed` in the markup.
- Local citations on the Indian directories that still matter: Justdial, Sulekha, IndiaMART, MagicBricks partner listings. Consistent NAP, no keyword stuffing in the business name, and no premises claims in any listing.

13. AI and assistant visibility

Assistants retrieve differently from crawlers, and this category is well suited to it: users ask conversational, comparative, high-stakes questions.

- Allow retrieval crawlers in robots.txt (see §6).
- Answer-first content structure: lead each section with the direct answer, then the detail. Assistants extract the first authoritative sentence.
- Publish comparison and pricing data in clean HTML tables, not images.
- Keep Organization markup complete and consistent, with sameAs links to the GBP, LinkedIn, and verified directory profiles.
- The guides section should be written to be cited, which means specific numbers, dated data, named authors with credentials, and clear sourcing.
- Consider a `/llms.txt` listing the canonical entry points. Support is uneven, the cost is an afternoon.

14. Launch checklist

Pre-launch

- All CI gates green: status codes, canonical parity, sitemap parity, duplication under 0.45, every page ≥ 3 inbound links
- `curl -I` confirms genuine 404 status on unknown paths; no templated 200s
- Uppercase, trailing-slash, www, and HTTP variants all 301 correctly
- Staging behind auth with `X-Robots-Tag: noindex`; confirm production does not carry it
- Structured data validates on one instance of every template
- Core Web Vitals green on the four heaviest templates
- Cache-Control headers verified per asset class; `cf-cache-status: HIT` on a second HTML request
- GSC and Bing Webmaster verified, sitemap index submitted
- Analytics, quote-start events, and call tracking firing per page type

Week 1

- Watch GSC Coverage daily for unexpected exclusions
- Confirm Googlebot crawl in Cloudflare logs; check nothing outside the manifest is being requested
- Verify the registered-office GBP resolves to the contact page

Ongoing

- Never ship a batch while the previous batch is below 60% indexed
- Re-run the publish gate quarterly; 301 the failures to their parents

15. Things that will go wrong

A short list of what actually breaks on projects like this, so nobody is surprised:

- Someone adds a catch-all route. Usually to "handle" a marketing campaign URL. It reintroduces the infinite URL space. Put a lint rule on `[...slug]` patterns.
- The manifest and the sitemap drift because someone adds a page family in Astro without adding it to the manifest. The CI parity check catches it; do not skip it under deadline pressure.
- Build time creeps past 20 minutes, engineers stop rebuilding, published data goes stale, and rate cards on the site stop matching what sales quotes. Watch build duration as a first-class metric.
- The duplication ceiling gets raised to ship one more batch. This is how every doorway farm started, including the one we audited.
- Cache purge is over-broad, every deploy cold-starts the whole edge cache, and TTFB spikes for an hour. Tag-based purging, tested.